Minimum period of distance-preserving syndrome extraction for bivariate-bicycle codes

Guancheng Pan[1, *]

[1] *Zhejiang University, Hangzhou, China*

(Dated: November 4, 2026)

Single-ancilla syndrome-extraction circuits for the gross code $[[144, 12, 12]]$ and the $[[90, 8, 10]]$ bivariate-bicycle code lose circuit distance unless overlapping X and Z checks are interleaved; how short a distance-preserving circuit can be was open. For translation-invariant circuits, in which all checks of one type share a CNOT timing repeated with period P, under circuit-level depolarizing noise, the shortest period preserving the code distance is 10 for the gross code and 11 for $[[90, 8, 10]]$, and we find the best circuit distance at every shorter period. The circuits work for every number of rounds, the impossibility results for sufficiently many. Interleaving acts through an integer cochain of round offsets: a circuit can be re-timed into a non-interleaved one exactly when this cochain is a coboundary, and a light extended-code logical is blocked when difference constraints along its hooks form a negative cycle. The price is idling: under standard idle noise the full-distance circuits are worse than IBM's depth-8 circuits at all directly resolvable error rates.

Bivariate-bicycle (BB) codes [1] are leading candidates for low-overhead quantum memory; IBM's architecture is built on the gross code $[[144, 12, 12]]$ [1, 2]. What a syndrome circuit delivers is its circuit distance d_{circ}, the minimum number of circuit faults that flip a logical observable without triggering a detector; at low physical error rates it governs the logical error rate. Single-ancilla circuits, in which each check is measured by one ancilla coupled to its data qubits by CNOTs, are the hardware-minimal choice, and the number of time steps per round, the period, sets how long every qubit idles. The design question is therefore the shortest period at which d_{circ} equals the code distance d.

Strikis, Browne and Beverland (SBB) [3] append to the check matrix the residual errors that ancilla faults spread onto the data and obtain an extended code of distance d_{ext}. Every single-ancilla circuit has $d_{\mathrm{circ}} \geq d_{\mathrm{ext}}$ (their Prop. 3), and a *non-interleaved* circuit, in which all X-type CNOTs precede all overlapping Z-type CNOTs or vice versa, has $d_{\mathrm{circ}} \leq d_{\mathrm{ext}}$ (their Cor. 1). For the gross code every choice of CNOT orders gives $d_{\mathrm{ext}} \leq 11$, so non-interleaved circuits are capped at 11 (their Prop. 4, by a numerical search over orders), and uniformly tiled ones at 10 (Prop. 5); for $[[90, 8, 10]]$ every choice of orders gives $d_{\mathrm{ext}} \leq 8$ (SM S13). IBM's depth-8 circuits have circuit distance at most 10 and 8 [1], values confirmed for fixed numbers of rounds [4, 5]; we show that they are exact for every number of rounds (Table I). Interleaved circuits can do better, but whether they can reach the full distance, and at what period, was open [3]. Known lower bounds on syndrome-circuit depth concern only the correctness of a single round: depth 6 is impossible and 7 is optimal for these codes [6–8]. Nothing was known about how short an interleaved circuit can be when it must also preserve distance, nor why interleaving helps at all.

We answer both questions within the natural class $\mathcal{T}(P)$ of translation-invariant single-ancilla circuits of period P (Definition 1). The minimum period with $d_{\mathrm{circ}} = d$ is exactly 10 for the gross code and 11 for $[[90, 8, 10]]$; at period 8 the best circuit distances are 11 and 8 (Theorem 1, Table I). The circuits reach these values for every number of rounds R and both memory bases; the impossibility results hold for sufficiently many rounds. The mechanism is an integer cochain δ that records, for every overlapping pair of checks, in which round one reads the errors the other deposits. A circuit can be re-timed into a non-interleaved one exactly when δ is a coboundary (Proposition 2); whether a light extended logical survives as an undetected fault set is a system of difference constraints, which a negative cycle along two hooks can make infeasible (Theorem 4, Corollary 5); and at period 8 a flat two-qubit hook costs one unit of distance because every pair of check qubits lies in a minimum-weight logical (Lemma 3).

Exact circuit distances and certified bounds exist for hypergraph-product codes [9], the smallest BB circuit [10] and other qLDPC constructions [11, 12]; morphing circuits [13, 14] and pipelined circuits [15] are other routes to short rounds, and interleaving X and Z checks recovers the full distance of the surface code at minimum depth [16]. Hooks aligned with minimum-weight logicals cost distance since the surface code [17], and check-dependent CNOT orders are the usual remedy [3, 18, 19]. For overlapping X and Z checks a single round is correct only if an even number of shared qubits see the X CNOT first [6–8]; δ is the integer, multi-round refinement of this $\mathbb{Z}_2$ parity.

Setting. We consider memory experiments in the Z or X basis: data initialisation, R rounds of syndrome extraction and transversal readout. Faults are the single-operation Pauli components of circuit-level depolarizing noise with idle errors (SD6 [20]); faults with identical detector and observable effects are merged into one detector-error-model mechanism [21], and d_{circ} counts mechanisms, minimised over both bases. In a Z-basis

TABLE I. Best circuit distance in $\mathcal{T}(P)$ for the gross code and $[[90,8,10]]$: upper bound for sufficiently many rounds (Theorem 1), attained for every R and both bases by the listed circuits (timings in Table II and Table S1). SBB's Fig. 12 circuit has period 8 and $d_{\mathrm{circ}} = 11$ but is not in $\mathcal{T}$.

P	gross	circuits (d_{circ})	$[[90,8,10]]$	circuits (d_{circ})
8	11	g_8 (11), IBM (10)	8	IBM (8)
9	11	g_9 (11)	9	b_9 (9)
10	12	g_{10} (12)	9	b_{10} (9)
11	12	u_{11} (12)	10	c_{11} (10)

memory only X errors matter: an error on an X-check ancilla after its k-th CNOT deposits the *residual* $\{q_{k+1},\dots,q_6\}$ of its order on the data (a *hook*), and the extended code appends the columns (Hs, Ls) of all residuals s to the data columns (H: check matrix, L: logicals); its distance is d_{ext} [3]. A BB code has checks $c \mapsto c + A_i$, $c + B_j$ $(i,j = 1,2,3)$, and every X check overlaps exactly nine Z checks $c + A_i + B_j$, on two qubits each (SM S3).

Definition 1 (class $\mathcal{T}(P)$) *A schedule is in $\mathcal{T}(P)$ if (i) every check has one ancilla, which is reset, coupled to its six data qubits by CNOTs and measured, in that order; (ii) all X checks share CNOT steps $t_X(\ell)$, $\ell \in \{A_1,\dots,B_3\}$, all Z checks share $t_Z(\ell)$, and round j adds jP; (iii) the six steps of each type are distinct and span at most $P-3$, leaving one step for reset and one for measurement; (iv) the six CNOTs on every data qubit fall on distinct steps modulo P; (v) the circuit measures the checks, i.e. on every overlapping pair of X and Z checks both shared qubits see the same relative round. CNOT orders are arbitrary and rounds may wrap around the period.*

Condition (v) is necessary for a deterministic circuit, $\mathcal{T}(P)$ is empty for $P < 8$, and IBM's depth-8 circuits lie in $\mathcal{T}(8)$.

Theorem 1 (optimal periods) *Under SD6 noise and for every sufficiently large number of rounds, every circuit in $\mathcal{T}(8)$ or $\mathcal{T}(9)$ has $d_{\mathrm{circ}} \le 11$ for the gross code, and every circuit in $\mathcal{T}(8)$ has $d_{\mathrm{circ}} \le 8$ and in $\mathcal{T}(9)$ or $\mathcal{T}(10)$ has $d_{\mathrm{circ}} \le 9$ for $[[90,8,10]]$. Each of these values, and $d_{\mathrm{circ}} = d$ at periods 10 (gross) and 11 ($[[90,8,10]]$), is attained in both memory bases for every $R \ge 1$ by the circuits of Table I.*

Hence, in $\mathcal{T}$, the minimum period with $d_{\mathrm{circ}} = d$ is 10 for the gross code and 11 for $[[90,8,10]]$.

The round-offset cochain. Fix a Z-basis memory. A data error at time u on qubit q is seen by Z check $z \ni q$ first in instance $\sigma_z(q,u) = \min\{t : \rho_z(q) + tP > u\}$, where $\rho_z(q)$ is the step at which instance 0 of z acts on q; it flips exactly the detector comparing instances $\sigma - 1$ and σ. A set of faults is undetected iff at every check the indices it produces pair up, and every SD6 fault is a data error, a hook or a readout flip of this staircase model (SM S2). An X-check ancilla error before the first CNOT deposits the whole check and is invisible, so for an X check c and a Z check z the two shared qubits receive the same index from instance 0 of c; we call it $\delta(c,z)$.

Proposition 2 (round-offset cochain) *Let every overlapping X and Z check share two qubits. (a) The instances can be relabelled so that every X instance acts on each shared qubit between two consecutive instances of each overlapping Z check iff $\delta(c,z) = b_z - a_c$ for integers a, b, i.e. δ is a coboundary. (b) Then every extended logical is realised by undetected faults for sufficiently many rounds, so $d_{\mathrm{circ}} \le d_{\mathrm{ext}}$. (c) If δ is constant in the round labelling of the memory circuit, (b) holds for every R; this is SBB's Cor. 1. For a circuit in $\mathcal{T}$, δ is a 3×3 matrix $\delta(A_i, B_j)$, a coboundary iff its nine entries are equal.*

Relabelling changes δ by a coboundary, which gives (a); for (b) each hook is taken in instance a_c and each data error just before its first X CNOT, so every check receives one common index an even number of times (End Matter). Interleaving in the cohomological sense is thus necessary for $d_{\mathrm{circ}} > d_{\mathrm{ext}}$ in the bulk, but it is not sufficient for $d_{\mathrm{circ}} = d$: IBM's gross circuit has a non-coboundary δ and $d_{\mathrm{circ}} = 10$ (Fig. 1a).

Short periods fail for a structural reason. For a hook residual S let $E(S)$ be the overlapping Z checks it meets an odd number of times; the hook is *flat* if δ is constant on $E(S)$.

Lemma 3 (pair obstruction) *Let a circuit be in $\mathcal{T}(P)$ and let S be the residual of a hook that leaves the last two qubits of an X-check order, or the last four (equal, modulo the check, to the first two). If S is flat and the two qubits lie in a weight-d logical, then for sufficiently many rounds $d-1$ faults form an undetected logical. For the gross code and for $[[90,8,10]]$ every pair of qubits of a check lies in a weight-d logical, for both check types.*

The hook replaces two data errors of the logical, and flatness lets all other data errors be read in the hook's round (Fig. 1c; the X basis is symmetric with the Z-check order). The general form costs $|L| - |T| + 1 + \min_v \sum_{e \in E(S)} |\delta(e) - v|$ faults (End Matter). A non-interleaved circuit in $\mathcal{T}$ has constant δ, so all its hooks are flat and Lemma 3 gives $d_{\mathrm{circ}} \le d - 1$ directly; within $\mathcal{T}$ this is the mechanism behind SBB's Prop. 4, there obtained by numerical search.

What decides is whether the light extended logicals can be realised. Fixing how each column of an extended logical S is realised (as a data error or a hook) and how the indices pair at each check, the hooks' instances x and the data errors' indices must satisfy constraints $x_a - x_b \le c_{ab}$: a hook contributes $x_c + \delta(c,z)$ at each check it meets

TABLE II. The full-distance circuits of minimum period. All checks of a type share the reset step, CNOT order, CNOT steps and measurement step (modulo P; a window passing the period boundary continues in the next period). Labels: $A = x^3+y+y^2$, $B = y^3+x+x^2$ (gross) and $A = x^9+y+y^2$, $B = 1+x^2+x^7$ ($[[90, 8, 10]]$) [1], $A \mapsto A_1A_2A_3$, $B \mapsto B_1B_2B_3$.

circuit	type	reset	CNOT order	CNOT steps	meas.
g_{10}, gross, $P = 10$	X	9	$A_1A_2B_1A_3B_2B_3$	0, 1, 2, 3, 4, 7	8
	Z	8	$B_3B_2A_3B_1A_2A_1$	9, 2, 3, 4, 5, 6	7
c_{11}, $[[90, 8, 10]]$, $P = 11$	X	0	$B_1B_3B_2A_3A_1A_2$	1, 2, 3, 4, 5, 9	10
	Z	10	$A_2A_1A_3B_2B_3B_1$	0, 4, 5, 6, 7, 8	9

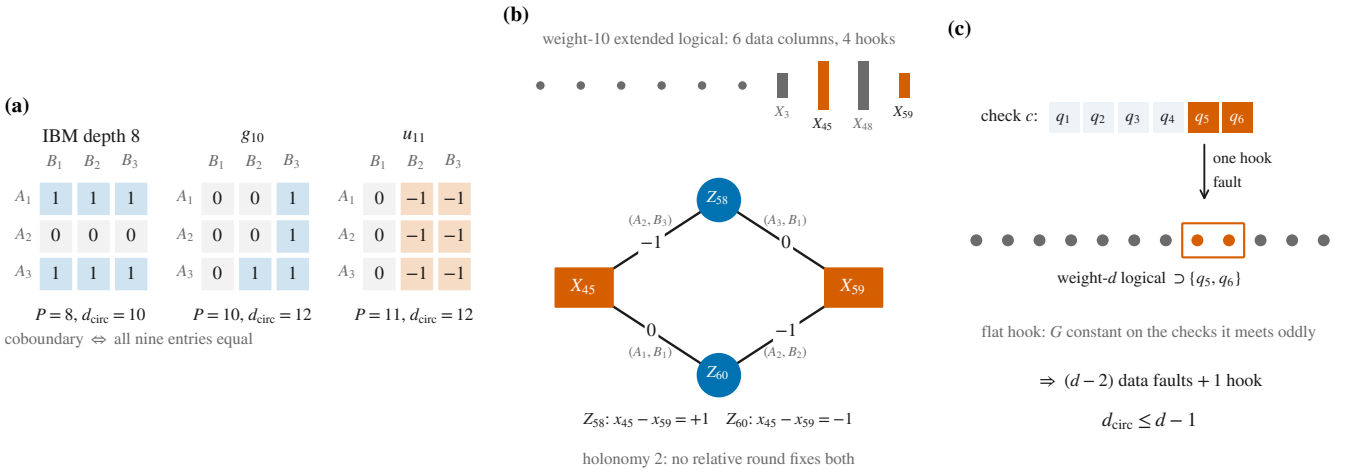


FIG. 1. How interleaving preserves distance. (a) The round-offset cochain $\delta(A_i, B_j)$ of three gross-code circuits in $\mathcal{T}$; it is a coboundary, i.e. the circuit can be re-timed into a non-interleaved one, only if all nine entries are equal. IBM's circuit shows that a non-coboundary δ is not enough for $d_{\text{circ}} = d$. (b) A negative cycle in u_{11} (Z memory). The first of its 144 weight-10 extended logicals contains hooks of X_{45} and X_{59} (residuals of length 4 and 2); Z_{58} and Z_{60} meet no other column. Edges carry the round index at which the reader sees the hook and the term pair. Cancelling at Z_{58} and at Z_{60} requires relative rounds differing by 2, so the logical cannot be realised; one readout flip shifts one constraint by 1 and does not help. Every weight-10 extended logical of u_{11} has such a certificate. (c) The pair obstruction (Lemma 3).

oddly, a data error any monotone staircase of drop at most one across its three readers. The window conditions require that each check's instance -1 reads before the first layer and instance R after the last; they hold for every circuit we use but not in general.

Theorem 4 (lifting criterion) *Fix a memory basis and a target $t \leq d$ with $t \leq d_{\text{ext}} + 2$. Assume the window conditions and that no hook residual has zero image. If, for every w with $d_{\text{ext}} \leq w \leq t-1$, no extended logical of weight w together with at most $t-1-w$ readout flips is realised by undetected faults in the bulk, then $d_{\text{circ}} \geq t$ for every $R \geq 1$.*

Corollary 5 (negative cycles) *For fixed realisations and pairings, realisability is infeasible iff the constraint graph has a negative cycle. In particular, if S contains hooks of X checks c, c' and two Z checks z, z' meet no other column of S, then S is not realised whenever $\delta(c, z) - \delta(c', z) \neq \delta(c, z') - \delta(c', z')$.*

The criterion is a finite, R-independent computation that turns SBB's qualitative observation that interleaving lets part of a propagating error be detected into a decidable statement. The two-hook certificate is a nonzero holonomy of δ around a 4-cycle of the overlap graph; Fig. 1(b) shows it for u_{11}, whose $\delta = -[j \neq B_1]$ blocks all 144 weight-10 extended logicals with holonomy 2. For the other attaining circuits some or all light logicals need the full difference system with data slack and pairing branches.

Proof of Theorem 1. A circuit in $\mathcal{T}(P)$ is described in the bulk by its signature, the two CNOT orders and δ; up to time translation and relabelling there are 22,608, 784,944 and 10,271,304 legal timing vectors at $P = 8$, 9 and 10, with 17,424, 67,536 and 267,984 signatures (SM S3).

Period 8. Every signature of $\mathcal{T}(8)$ has a flat pair hook, the first or last pair of the X or Z order, so Lemma 3 gives $d_{\text{circ}} \leq d-1$ for both codes; an explicit weight-$(d-1)$ witness was constructed for every legal timing vector (SM S6). For $[[90, 8, 10]]$ the stronger bound 8 comes from the patterns below.

Periods 9 and 10. Explicit undetected fault patterns,

a pair hook with re-timed data errors or several hooks lifting a light extended logical, cover every legal timing vector of $\mathcal{T}(8)$ and $\mathcal{T}(9)$ for both codes and of $\mathcal{T}(10)$ for $[[90, 8, 10]]$, with weights 11 (gross) and 8, 9, 9. Each pattern's residual is a nontrivial logical, and each is confirmed by injection into the memory circuit. Coverage is certified twice: the satisfiability encodings are refuted by CaDiCaL [22] with DRAT proofs checked by drat-trim [23], and an exhaustive check without SAT runs over all canonical timing vectors. In sampled cases the witnesses are already undetected for $R \geq 2$ (SM S7).

Attaining circuits. For g_{10} (Table II) Theorem 4 applies with $t = 12 = d_{\text{ext}} + 2$: the window conditions hold, no hook has zero image, and none of the 3,204 weight-10 extended logicals (with at most one readout flip) and 121,104 of weight 11 is realised in the bulk, in either basis; a weight-12 data logical applied after initialisation gives the upper bound. An exhaustive search of the full detector error model, which uses neither the extended code nor the criterion, finds no undetected logical of weight ≤ 11 for $R = 1$, 2, 3 in both bases. c_{11} is treated in the same way ($d_{\text{ext}} = 8$, lists of 585 and 16,830), and a second proof combines exact lifts for $R \leq 25$ with a gap argument (End Matter). Every infeasibility claim used in the proofs for g_{10}, c_{11} and the impossibility results is accompanied by a checked certificate (VeriPB [24, 25] for d_{ext} and list completeness, DRAT with LRAT cores checked by cake_lpr [26] for the lifts); the encodings are generated by our code and cross-checked against independent implementations, and the reductions are the mathematical arguments given here, whose premises were verified computationally. The remaining circuits of Table I satisfy the criterion with $t \leq d_{\text{ext}} + 2$ by two independent solvers and a direct search for $R \leq 2$, 3 or 4; their upper bounds are explicit witnesses at $R = 1$ that persist to every R (End Matter). In particular IBM's circuits have $d_{\text{circ}} = 10$ and 8 for every R, and g_8 exceeds them by one at the same period, equal to SBB's non-interleaved Fig. 12, which lies outside $\mathcal{T}$.

Optimality within a signature. A signature is realisable at period P iff a difference system has no positive cycle; the signatures of u_{11} and c_{11} have minimum period exactly 11 (End Matter). The gain at $P = 10$ therefore comes from a different offset pattern (δ of g_{10} in Fig. 1a), not from compressing the period-11 one. □

Cost. The extra steps of the period are idle steps: two per ancilla and round for g_{10}, three for c_{11}. Under SD6 with idle noise and BP-OSD decoding, the full-distance circuits have a higher per-round logical error rate than IBM's depth-8 circuits throughout the directly resolvable range: for the gross code by a factor of about 5–10 at $p = 2\times10^{-3}$ and at least about 1.5 down to $p = 1.5\times10^{-3}$ (g_{10} itself is about 2.3 times below u_{11} at 2×10^{-3}), and for $[[90, 8, 10]]$ at all $p \geq 10^{-3}$. Where, below 10^{-3}, they would overtake IBM's circuits depends on the smallest number of faults at which BP-OSD fails on each circuit, which our data do not resolve; plausible extrapolations place the crossover anywhere from about 10^{-3} to below 10^{-4}, or rule it out. In 23 of 24 sampled failures BP-OSD returned a correction less likely than the actual fault set, so at these error rates the comparison is between circuit–decoder pairs rather than distances. Per round, IBM's $[[90, 8, 10]]$ circuit has exactly 315 (Z) and 540 (X) undetected logicals of weight 8, while c_{11} has none below weight 10. When idling is free ($p_{\text{idle}} = 0$) the gain is visible: c_{11} has a lower per-round logical error rate than IBM's circuit at every sampled p from 1.2×10^{-3} (ratio 0.51 [0.31, 0.83], BP-OSD) to 5×10^{-3} (with both BP-OSD and Relay-BP from 2×10^{-3}; 0.52 [0.44, 0.62] at 3×10^{-3}), while at $p_{\text{idle}} = 0.1p$ the two are indistinguishable. A neutral-atom model with literature parameters gives c_{11} the lower rate (ratios 0.40–0.82) provided the two ancilla arrays move one after the other and atoms left unpaired during entangling pulses do not err; with error on unpaired atoms and simultaneous moves the ordering reverses (SM S11).

Discussion. Within translation-invariant single-ancilla circuits with one CNOT timing per check type, separate reset and measurement steps, and SD6 noise, the minimum distance-preserving period of both codes is now known, and so is the best distance at every shorter period. The lower bounds come from explicit fault patterns, and period 8 is excluded by one structural property, the pair covering of Lemma 3. The upper bounds come from a criterion that is local in time and global over the light extended logicals. The same period-11 timing template reaches $d_{\text{circ}} = 10$ on $[[108, 8, 10]]$ but fails on $[[90, 8, 10]]$ with the same labels, so term labels must be chosen per code (SM S12).

Four questions remain open. Can per-check timings, outside $\mathcal{T}$, reach full distance at period 8 or 9? Can merged measure-and-reset steps, which shorten every period by one, do the same? Does a criterion like Theorem 4 hold for $d \geq d_{\text{ext}} + 3$, needed for larger codes, and are the window conditions necessary? And where does the logical error rate of a full-distance circuit cross that of a shorter one under a decoder whose low-weight failures are resolved? An analytic proof of the bounds at periods 9 and 10 would also replace the computer-assisted part of Theorem 1.

Data availability. Circuit specifications, lists of extended logicals, fault patterns, witnesses, certificates and reproduction scripts are available at [Zenodo DOI].

AI use. This research was carried out with AI agents (Claude, Anthropic). The agents proposed hypotheses, wrote and ran the search and verification code, worked out the constructions and proofs, and independently rechecked each claim reported here, using separately written code for every computation. The author directed the project, made all decisions at its decision points, and takes full responsibility for the content. In line with APS policy, no AI system is listed as an author.

* 3250101086@zju.edu.cn

[1] S. Bravyi, A. W. Cross, J. M. Gambetta, D. Maslov, P. Rall, and T. J. Yoder, High-threshold and low-overhead fault-tolerant quantum memory, Nature **627**, 778 (2024), arXiv:2308.07915.

[2] T. J. Yoder, E. Schoute, P. Rall, E. Pritchett, J. M. Gambetta, A. W. Cross, M. Carroll, and M. E. Beverland, Tour de gross: A modular quantum computer based on bivariate bicycle codes, arXiv:2506.03094 (2025).

[3] A. Strikis, D. E. Browne, and M. E. Beverland, High-performance syndrome extraction circuits for quantum codes, arXiv:2603.05481 (v2, 2026).

[4] L. Aghababaie Beni, O. Higgott, and N. Shutty, Tesseract: A search-based decoder for quantum error correction, arXiv:2503.10988 (v2, 2025).

[5] C. Crocker, The gross code's syndrome circuit has circuit-level distance 10: an exhaustive lower bound, Zenodo (2026), doi:10.5281/zenodo.23452754.

[6] G. Li, Y. Gan, Z. Wu, X. Zhang, Z. Zhang, and J. Liu, A parity obstruction to degree-optimal syndrome extraction in bivariate bicycle codes, arXiv:2509.26494 (v4, 2026).

[7] K. Zhang, D. Gao, Z. Yang, R. Zhou, F. Liu, Z. Ji, and J. Chen, Optimal compilation of syndrome extraction circuits for general quantum LDPC codes, arXiv:2603.21499 (2026).

[8] E. Durazo Rocha, O. Å. Mostad, H.-Y. Lin, and E. Rosnes, Local automorphism-aware syndrome compilation for general quantum LDPC codes, arXiv:2609.40319 (2026).

[9] A. G. Manes and J. Claes, Distance-preserving stabilizer measurements in hypergraph product codes, Quantum **9**, 1618 (2025), arXiv:2308.15520.

[10] M. Webster, A. Jacob, and O. Higgott, Distance-finding algorithms for quantum codes and circuits, arXiv:2603.22532 (v2, 2026).

[11] Z. Liang, B. Gu, Y.-A. Chen, J. Eisert, and Z. Wang, Ultra-high-distance quantum memories from amplified qLDPC codes, arXiv:2609.37231 (2026).

[12] S. Ghosh, H. Boche, and A. Tanggara, Fault tolerance of quantum circuits with tensor networks and symplectic geometry, arXiv:2610.03429 (v3, 2026).

[13] M. H. Shaw and B. M. Terhal, Lowering connectivity requirements for bivariate bicycle codes using morphing circuits, Phys. Rev. Lett. **134**, 090602 (2025), arXiv:2407.16336.

[14] A. Wills, M. E. Beverland, L. S. Bishop, J. M. Gambetta, P. Rall, V. Siddhu, and A. W. Cross, Concatenating algebraic codes over high-rate quantum LDPC codes, arXiv:2605.21898 (v2, 2026).

[15] Q. Xu, J. P. Bonilla Ataides, C. A. Pattison, N. Raveendran, D. Bluvstein, J. Wurtz, B. Vasić, M. D. Lukin, L. Jiang, and H. Zhou, Constant-overhead fault-tolerant quantum computation with reconfigurable atom arrays, Nat. Phys. **20**, 1084 (2024), doi:10.1038/s41567-024-02479-z, arXiv:2308.08648.

[16] Y. Hirai, S. Ikari, Y. Ueno, and Y. Suzuki, No more hooks in the surface code: Distance-preserving syndrome extraction for arbitrary layouts at minimum depth, arXiv:2603.01628 (v2, 2026).

[17] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill, Topological quantum memory, J. Math. Phys. **43**, 4452 (2002), arXiv:quant-ph/0110143.

[18] Y. Tomita and K. M. Svore, Low-distance surface codes under realistic quantum noise, Phys. Rev. A **90**, 062320 (2014), arXiv:1404.3747.

[19] G. Kishony and A. Fowler, Color code off-the-hook: avoiding hook errors with a single auxiliary per plaquette, arXiv:2603.28852 (2026).

[20] C. Gidney, M. Newman, A. Fowler, and M. Broughton, A fault-tolerant honeycomb memory, Quantum **5**, 605 (2021), arXiv:2108.10457.

[21] C. Gidney, Stim: a fast stabilizer circuit simulator, Quantum **5**, 497 (2021), arXiv:2103.02202.

[22] A. Biere, T. Faller, K. Fazekas, M. Fleury, N. Froleyks, and F. Pollitt, CaDiCaL 2.0, in *Computer Aided Verification (CAV 2024)* (Springer, Cham, 2024), pp. 133–152, doi:10.1007/978-3-031-65627-9_7.

[23] N. Wetzler, M. J. H. Heule, and W. A. Hunt, Jr., DRAT-trim: Efficient checking and trimming using expressive clausal proofs, in *Theory and Applications of Satisfiability Testing (SAT 2014)* (Springer, Cham, 2014), pp. 422–429, doi:10.1007/978-3-319-09284-3_31.

[24] B. Bogaerts, S. Gocht, C. McCreesh, and J. Nordström, Certified dominance and symmetry breaking for combinatorial optimisation, J. Artif. Intell. Res. **77** (2023), doi:10.1613/jair.1.14296.

[25] S. Gocht and J. Nordström, Certifying parity reasoning efficiently using pseudo-Boolean proofs, Proc. AAAI Conf. Artif. Intell. **35**, 3768 (2021), doi:10.1609/aaai.v35i5.16494.

[26] Y. K. Tan, M. J. H. Heule, and M. O. Myreen, cake_lpr: Verified propagation redundancy checking in CakeML, in *Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2021)*, Lecture Notes in Computer Science Vol. 12652 (Springer, Cham, 2021), pp. 223–241, doi:10.1007/978-3-030-72013-1_12.

[27] See Supplemental Material at [URL will be inserted by publisher] for the class $\mathcal{T}(P)$, the staircase model, the cochains of all circuits, the proofs of the impossibility results and of the attaining circuits, the certificates, the logical error rates, the neutral-atom noise model, a transfer example and earlier constructions, which includes Refs. [1, 3, 15, 20–26, 28–34].

[28] J. Viszlai, W. Yang, S. F. Lin, J. Liu, N. Nottingham, J. M. Baker, and F. T. Chong, qSIEVE: Efficient qLDPC memory via systolic movement in atom arrays, ACM Trans. Quantum Comput. **7**, 9 (2026), arXiv:2311.16980.

[29] S. J. Evered, D. Bluvstein, M. Kalinowski, S. Ebadi, T. Manovitz, H. Zhou, S. H. Li, A. A. Geim, T. T. Wang, N. Maskara, H. Levine, G. Semeghini, M. Greiner, V. Vuletić, and M. D. Lukin, High-fidelity parallel entangling gates on a neutral atom quantum computer, Nature **622**, 268 (2023), arXiv:2304.05420.

[30] D. Bluvstein *et al.*, Logical quantum processor based on reconfigurable atom arrays, Nature **626**, 58 (2024), doi:10.1038/s41586-023-06927-3, arXiv:2312.03982.

[31] D. Bluvstein *et al.*, A fault-tolerant neutral-atom architecture for universal quantum computation, Nature **649**, 39 (2026), arXiv:2506.20661.

[32] J. A. Muniz *et al.*, High-fidelity universal gates in the ^{171}Yb ground-state nuclear-spin qubit, PRX Quantum **6**, 020334 (2025), arXiv:2411.11708.

[33] A. G. Radnaev *et al.*, Universal neutral-atom quantum computer with individual optical addressing and non-destructive readout, PRX Quantum **6**, 030334 (2025), arXiv:2408.08288.

[34] R. B.-S. Tsai, X. Sun, A. L. Shaw, R. Finkelstein, and M. Endres, Benchmarking and fidelity response theory of high-fidelity Rydberg entangling gates, PRX Quantum **6**, 010331 (2025), arXiv:2407.20184.

END MATTER

Proof of Proposition 2. Relabelling instance j of X check c as $j + a_c$ and of Z check z as $j + b_z$ changes δ to $\delta - (b_z - a_c)$, and $\delta \equiv 0$ is the statement of (a). If $\delta \equiv 0$, every X CNOT on q lies in $(\rho_z(q) - P, \rho_z(q))$ for every reader z, so a data error just before the first of them has index 0 at all readers; taking hooks in instance a_c, every check z receives index b_z exactly $(HS)_z$ times, an even number. If δ is constant in the round labelling of the memory circuit, all these faults lie in one round and exist for every R, which is (c). A translation-invariant coboundary is constant after summing over translations. □

Proof of Lemma 3. Let $T \in \{S, S^c\}$ be the pair inside L and v a value minimising $\sum_{e \in E(S)} |\delta(e) - v|$. Place every qubit of $L \setminus T$ after its last read, all at one index s_0, and the hook in the instance with index s_0 at value v. A check outside $E(S)$ receives 0 or 2 equal hook indices and an even number of data indices s_0; a check $e \in E(S)$ receives one hook index and an odd number of s_0, mismatched by $|\delta(e) - v|$, which as many readout flips repair. The residual is L or L plus the check, a nontrivial logical, with $|L| - |T| + 1 + \sum |\delta(e) - v|$ faults; for a flat pair hook and $|L| = d$ this is $d - 1$. □

Light logicals. Let an undetected logical consist of w faults, z of zero image and c pairs of equal nonzero images. Summing all detectors of each check telescopes to the check's parity on the final readout, so the remaining $w - z - 2c$ distinct columns form an extended logical S, and $|F| = |S| + z + 2c$ with $|S| \geq d_{\text{ext}}$. For $w \leq d_{\text{ext}} + 1$ this forces $c = 0$ and $z \leq w - d_{\text{ext}}$. Without zero-image hooks the zero-image faults are readout flips.

Window lemma. The finite index of a deposit is the number of reads of instances $0, \ldots, R-1$ before it (R for the final readout). If instance -1 of every check reads before the first layer and instance R after the last, the finite index equals the bulk index for every deposit inside the circuit. Hence every undetected set at finite R is an undetected bulk set with the same columns (inclusion only: the bulk may use instances outside $0, \ldots, R-1$).

Proof of Theorem 4. An undetected logical F with $|F| \leq t - 1 \leq d_{\text{ext}} + 1$ at some R embeds in the bulk by the window lemma; by the count above it realises an extended logical of weight $w \in [d_{\text{ext}}, t-1]$ with at most $t - 1 - w$ readout flips, which the hypothesis excludes. □

Proof of Corollary 5. A hook of instance x_c contributes $x_c + \delta(c, z)$ at each check $z \in E(S)$; a data error on q contributes indices that, ordered cyclically by read time, satisfy $y_1 \geq y_2 \geq y_3 \geq y_1 - 1$; equal indices at a check are paired, and a readout flip adds $\{s, s+1\}$. All constraints are of the form $x_a - x_b \leq c_{ab}$, feasible iff no negative cycle exists. At a check met only by hooks c, c', pairing forces $x_c - x_{c'} = \delta(c', z) - \delta(c, z)$; two such checks with different right sides give a cycle of weight $-|h|$, h the holonomy. With $|h| \geq 2$ one readout flip cannot close it. □

Certificates for g_{10}. $d_{\text{ext}} = 10$ and the completeness of the lists of 3,204 weight-10 and 121,104 weight-11 extended logicals are certified in both bases by VeriPB proofs. Bulk realisability reduces to a finite instance by a compression lemma: in the round labelling of the memory circuit every fault of g_{10} spans at most one index; in an undetected realisation of at most 11 faults, equal indices at a check link faults into components, each component can be shifted by whole rounds independently, and here only one component exists (otherwise a part would be a lighter logical or trivial). Its indices span at most 11 and shift into $[1, 12]$, which avoids the first-round and final detectors of the circuit with $R^* = 13$ rounds; the window lemma then makes the finite and bulk indices equal. At $R = 13$ none of the 248,616 list items (both bases, no symmetry reduction) is realised, certified by DRAT proofs checked with drat-trim and, for their LRAT cores, cake_lpr.

Gap argument for c_{11}. Every check instance of c_{11} starts at an offset 0 or 1 modulo $D = 11$, is measured 10 layers later and has its last CNOT at offset 9. If all faults of a set lie at layers $\leq a$, the data frame they leave is constant from layer $a + D - 1$ on; a detector responds to faults before a and after b only if $b - a \leq 30$. A fault-free window of 31 layers therefore splits an undetected logical into two undetected parts, one of them lighter. With no undetected logical of weight $\leq w - 1$, one of weight w spans at most $30(w-1)$ layers. The circuit with R rounds equals the circuit with $R-1$ rounds with an 11-layer block inserted at any layer $x \in [1, 11(R-1)]$, so deleting fault-free blocks maps every undetected logical of weight 8 (9) into a circuit with at most 22 (25) rounds. Exact lifts of the 585 and 16,830 extended logicals for $R \leq 25$, both bases, are zero and certified by DRAT proofs; $d_{\text{ext}} \geq 8$ and the list completeness are certified by VeriPB.

Persistence of upper bounds. The memory circuit with $R \geq R_0$ rounds contains the circuit with R_0 rounds as its first layers (the start offsets do not depend on R), and the extra gates there are later instances of the depositing checks, whose CNOTs move no data error. If instance R_0 of every reading check reads after the last layer of the R_0-round circuit, an undetected logical of that circuit stays undetected with the same logical effect at every $R \geq R_0$: the data error after the last fault is unchanged, every later comparison sees the same syndrome, and the final readout agrees. This condition is independent of R and

holds for g_8, g_9, b_9, b_{10} and IBM's circuits, so their $R = 1$ witnesses of weight 11, 11, 9, 9, 10 and 8 bound d_{circ} for every R; for g_{10}, u_{11} and c_{11} a weight-d data logical applied after initialisation suffices. More generally, for translation-invariant schedules in which all checks of each type share one preparation time, the detector error model of R rounds embeds into that of $R + 1$ rounds, in place or delayed by one period, so the circuit distance is non-increasing in R; we know of no schedule whose circuit distance increases with R.

Period lemma. A signature is realisable at period P iff the integer system $x_{\pi(i+1)} - x_{\pi(i)} \geq 1$, $x_{\pi(6)} - x_{\pi(1)} \leq P - 3$ for both orders and $G(k,l)P + 1 \leq t_X(A_k) - t_Z(B_l) \leq G(k,l)P + P - 1$, likewise on the right qubit (G equals δ up to a constant, SM S3), has an integer solution, i.e. its constraint graph has no positive cycle. For u_{11} the longest chain through a check window has eight steps, so its minimum period is $8 + 3 = 11$; the same holds for c_{11}, and neither signature occurs at $P = 8$, 9 or 10.

Supplemental Material

The sections follow the main text: S1 notation and noise, S2 the staircase model, S3 the class $\mathcal{T}(P)$, S4 the cochains of all circuits, S5 the criterion in practice, S6 period 8, S7 periods 9 and 10, S8 the attaining circuits, S9 certificates, S10 logical error rates, S11 the neutral-atom model, S12 a transfer example, S13 earlier constructions, S14 verification and data.

S1. NOTATION AND NOISE MODEL

Codes. The gross code [1] has $\ell = 12$, $m = 6$, $A = x^3 + y + y^2$, $B = y^3 + x + x^2$, $H_X = [A \,|\, B]$, $H_Z = [B^T \,|\, A^T]$; qubits and checks are indexed by $i = 6a + b$ for the monomial $x^a y^b$, with left and right blocks of 72 qubits. Term labels: $A_1 = x^3$, $A_2 = y$, $A_3 = y^2$, $B_1 = y^3$, $B_2 = x$, $B_3 = x^2$. The X check at (x, y) acts on left qubit $(x, y) + A_k$ and right qubit $(x, y) + B_k$; the Z check at (x, y) on left qubit $(x, y) - B_k$ and right qubit $(x, y) - A_k$. $[[90, 8, 10]]$ has $\ell = 15$, $m = 3$, $A = x^9 + y + y^2$, $B = 1 + x^2 + x^7$ [1], labels $A_1 = x^9$, $A_2 = y$, $A_3 = y^2$, $B_1 = 1$, $B_2 = x^2$, $B_3 = x^7$, index $i = 3a + b$; an exhaustive search finds no logical of weight ≤ 9, so $d = 10$.

Circuits. In every round each ancilla is reset, applies its six CNOTs (ancilla as control for X checks, as target for Z checks) and is measured. A Z-basis memory initialises the data in $|0\rangle$, runs R rounds and reads all data in the Z basis; detectors are the first measurement of each Z check, consecutive comparisons, and the comparison of the last measurement with the final readout. The X basis is analogous.

Noise and d_{circ}. SD6 [20]: a two-qubit depolarizing channel after every CNOT, a flip after every reset and before every measurement, a single-qubit depolarizing channel on every qubit idle in a time step. A fault is one Pauli component of one channel; faults with identical detector and observable effects form one mechanism, and d_{circ} counts mechanisms. In a Z-basis memory only Z-check detectors exist, so only the X part of each fault matters. d_{circ} of a circuit is the smaller of the values of the two bases; all circuits we construct reach it in both. In the logical-error-rate studies the idle channel is rescaled, $p_{\mathrm{idle}} = \lambda p$.

S2. THE STAIRCASE MODEL

Take a Z-basis memory. Writers are X checks: an X error on the ancilla after its $(k-1)$-th CNOT is deposited on $q_k, \ldots, q_6$ at the times of their CNOTs. Readers are Z checks; instance t of z reads q at $\rho_z(q) + tP$. The index of a deposit at time u is $\sigma_z(q, u) = \min\{t : \rho_z(q) + tP > u\}$. Measurement m_t of z equals the parity of deposits with index $\leq t$ plus its own readout flip, so a deposit of index s flips exactly detector $m_s + m_{s-1}$ (the final detector for $s = R$), and a readout flip of instance t acts as the two indices $\{t, t+1\}$. A fault set is undetected iff every check receives each index an even number of times; its logical effect is the L image of the remaining data error. Every SD6 fault component is one model fault (Table S1); conversely every model fault is an SD6 fault, since every data qubit idles or takes part in a CNOT in every layer. For all circuits used here the set of model signatures equals the set of detector signatures of the SD6 detector error model for $R = 1$, 2 (and 3 where checked), in both bases.

TABLE S1. SD6 fault components in a Z-basis memory and the model faults they are.

location and X part	model fault
after the k-th CNOT of an X check: X_a	hook with residual $q_{k+1}, \ldots, q_6$
same: X_q / $X_a X_q$	data error on q_k after the CNOT / hook with residual $q_k, \ldots, q_6$
after a CNOT of a Z check (data q, ancilla b): X_q / X_b / $X_q X_b$	data error after the read / readout flip / data error before the read
data idle, after data reset, before final readout	data error at that time
X-check ancilla after reset, before the first CNOT	whole check deposited: no signature
Z-check ancilla after reset, idle, before measurement	readout flip

Images and light logicals. Each fault's data image is zero, one data column, or one residual column; the image of a whole check is zero. Augmenting the circuit with one observable per row of H_Z and L_Z on the final readout, the sum of all detectors of a check telescopes to its row parity, so the nonzero images of an undetected fault set sum to

an extended logical (End Matter). For all circuits used, the zero-image mechanisms are exactly the readout flips: no hook residual with $k = 2, \ldots, 6$ has zero image.

S3. THE CLASS $\mathcal{T}(P)$

Overlaps. For both codes the nine sums $A_i + B_j$ are distinct, so X check c overlaps exactly the nine Z checks $c + A_i + B_j$, each on two qubits: the left qubit $c + A_i$ (term A_i for the X check, B_j for the Z check) and the right qubit $c + B_j$ (term B_j for X, A_i for Z); 648 overlapping pairs for the gross code and 405 for $[[90, 8, 10]]$.

Determinism. Let $G_L(i,j) = \lfloor (t_X(A_i) - t_Z(B_j))/P \rfloor$ and $G_R(i,j) = \lfloor (t_X(B_j) - t_Z(A_i))/P \rfloor$. X instance r acts on the left shared qubit before Z instance s iff $s - r > G_L$, and likewise on the right. The measured operator is the check iff the number of shared qubits on which X acts first is even for every $s - r$, i.e. $G_L = G_R =: G$. Schedules violating this produce non-deterministic detectors in the simulator. A hook of instance r is seen by the Z check $c + A_i + B_j$ at index $r + G(i,j)$ plus a constant fixed by the instance labelling, so δ and G differ by a constant; flatness and costs are the same for both.

Signatures. The bulk fault model of a circuit in $\mathcal{T}(P)$ depends only on its signature (π_X, π_Z, δ): data errors depend on the read order of the three Z checks of a qubit, hooks on the suffixes of π_X and on δ, and readout flips on nothing. Canonical form: $\min t_X = 0$ and $0 \le t_X(A_1) - t_Z(B_1) \le P - 1$; every legal schedule has exactly one canonical timing vector, by a global time shift and a relabelling of Z instances.

TABLE S2. Canonical legal timing vectors and signatures of $\mathcal{T}(P)$.

P	8	9	10
timing vectors	22,608	784,944	10,271,304
signatures	17,424	67,536	267,984

Two independent enumerations give these counts. For $P < 8$ six distinct steps cannot fit in a span of $P - 3$, so $\mathcal{T}(P)$ is empty.

S4. THE COCHAINS OF ALL CIRCUITS

Table S3 lists $\delta(A_i, B_j)$ in the Z basis (rows A_1, A_2, A_3, columns B_1, B_2, B_3); in the X basis δ changes by a constant. None is a coboundary, so none of these circuits can be re-timed into a non-interleaved one. u_{11} and c_{11} carry a single-term twist, on B_1 and on A_2; IBM's circuits carry one on A_2 of the opposite sign to c_{11} and stop at $d_{\mathrm{circ}} = 10$ and 8, so the twist must match the light extended logicals of the order class. g_8 and b_9 are twisted on A_3, and g_9, g_{10}, b_{10} carry two-valued patterns that are not single-term twists. A non-interleaved control with the orders of u_{11} has $\delta \equiv 0$ and realises all 144 weight-10 extended logicals in the bulk.

TABLE S3. $\delta(A_i, B_j)$ of the circuits of Table I (rows A_i, columns B_j), from the specifications.

circuit	δ	circuit	δ
IBM gross, $P = 8$	$(1,1,1;\ 0,0,0;\ 1,1,1)$	IBM $[[90,8,10]]$, $P = 8$	$(1,1,1;\ 0,0,0;\ 1,1,1)$
g_8	$(1,1,1;\ 1,1,1;\ 2,2,2)$	b_9	$(1,1,1;\ 2,1,1;\ 2,2,2)$
g_9	$(0,0,1;\ 0,1,1;\ 0,1,1)$	b_{10}	$(1,0,0;\ 1,0,0;\ 1,0,1)$
g_{10}	$(0,0,1;\ 0,0,1;\ 0,1,1)$	c_{11}	$(-1,-1,-1;\ 0,0,0;\ -1,-1,-1)$
u_{11}	$(0,-1,-1;\ 0,-1,-1;\ 0,-1,-1)$		

S5. THE CRITERION IN PRACTICE

Window conditions. They reduce to R-independent inequalities, since the start offsets of the memory circuit do not depend on R. They hold, with finite and bulk indices equal for every deposit, for all circuits of Table I (checked for

$R = 1, 2, 3$ for every circuit, and up to $R = 12$ or 20 for most). They fail for most random legal schedules (61 of 80 schedule-basis pairs in a random sample), so they must be checked per circuit; we know no example in which their failure creates additional finite-R lifts.

Solvers. Bulk realisability was decided by a CP-SAT model (one integer shift per realisation, perfect matchings of equal indices at each check, readout flips as two extra items) and by a Bellman–Ford solver of the difference system with pairing branches, each written independently; every reported realisation was re-checked by counting index parities. Positive controls: non-interleaved controls with the same orders realise all listed logicals; u_{16}, an interleaved $[[90, 8, 10]]$ circuit with $d_{\mathrm{circ}} = 9$, realises 315 of its weight-9 logicals in both bases.

Short certificates. Two-hook (or longer hook-only) cycles certify all 144 weight-10 logicals of u_{11} (holonomy 2), 5,400 of its 7,344 of weight 11, and 405 of the 585 weight-8 logicals of c_{11}; the rest need data slack or pairing branches. For g_{10} only 1,224 of 3,204 weight-10 logicals have a two-hook certificate, all with holonomy 1, and none of the 432 weight-9 logicals of g_8: their non-realisability rests on exact solving.

S6. PERIOD 8

Pair table. For both codes and both check types, each of the 15 pairs of term labels lies in a weight-d logical (witnesses verified by linear algebra, 60 of 60). For triples, 8 ($[[90, 8, 10]]$) and 10 (gross) of the 20 lie in weight-d logicals.

Flat pair hooks. The pair hooks of an order π are the last pair $S_2 = \{\pi_5, \pi_6\}$ and the residual of length 4, which equals the first pair $\{\pi_1, \pi_2\}$ modulo the check. Among the 17,424 signatures of $\mathcal{T}(8)$, every one has at least one flat pair hook among those of π_X and π_Z (an exhaustive check). For each of the 22,608 legal timing vectors and both codes we constructed the weight-$(d-1)$ witness of Lemma 3 explicitly (the pair hook, the remaining $d-2$ qubits of the weight-d logical after their last read, hook instance between -4 and 4) and verified it in the bulk model; sampled witnesses were replayed in the memory circuit by stim.

S7. PERIODS 9 AND 10

Patterns. A pattern is a set of data errors (residual qubits with read-order cuts), hooks (check type, suffix, round offset) and readout flips, together with the timing conditions under which it is undetected in the bulk model. The five results use 482 ($[[90, 8, 10]]$, $P = 8$, weight ≤ 8), 856 ($P = 9$, ≤ 9), 1,718 ($P = 10$, ≤ 9), 599 (gross, $P = 8$, ≤ 11) and 1,186 (gross, $P = 9$, ≤ 11) patterns. Every pattern's residual data error lies in $\ker H$ and outside the row space of the opposite check matrix, i.e. is a nontrivial logical (4,841 of 4,841).

Certificates. For each result a CNF encodes legality (one-hot steps, distinctness, canonical form, span, data-qubit distinctness modulo P, determinism) and one clause per pattern forbidding the timings under which it applies. Each of the five formulas is unsatisfiable; CaDiCaL produces DRAT proofs verified by drat-trim. Only time shift and Z-instance relabelling are used as symmetry reductions; neither changes the bulk physics.

SAT-free check. Independently, every canonical legal timing vector (all 10,271,304 at $P = 10$) was tested against the patterns with a separately written bulk model: none is uncovered. The bulk model agrees with stim on 240 of 240 random fault sets, and every pattern used was injected into the memory circuit on the vectors it covers (none of 10,293 injections failed; removing or shifting one fault always triggered a detector).

Small R. The results are bulk statements. For 25 random vectors per result the covering pattern is already undetected, after a global instance shift, in the memory circuit with $R = 1$ or 2, and remains so for the next three R.

S8. THE ATTAINING CIRCUITS

All circuits pass the simulator's legality checks, are deterministic, measure every check on its own support (single-qubit injections at every layer boundary), and are invariant under all translations. Table S5 summarises the lower and upper bounds.

Lists. Extended-code lists were enumerated by a depth-first search branching on the lowest unsatisfied check and pruning at zero syndrome, complete for weights below $d_{\mathrm{ext}} + 3$, with orbit representatives under the translations expanded and re-verified; for every circuit a second, independently written enumeration gives identical sets.

Criterion. Target $t = d$ for g_{10}, u_{11}, c_{11}, and $t = d_{\mathrm{circ}}$ for the others; in every case $t \leq d_{\mathrm{ext}} + 2$. Bulk realisability was decided on every list item by CP-SAT, and on orbit representatives by the Bellman–Ford solver, with zero realisations

TABLE S4. Timings of the other circuits of Table I (conventions of Table II).

circuit	type	reset	CNOT order	CNOT steps	meas.
IBM gross, $P = 8$	X	7	$A_2B_2B_1B_3A_1A_3$	0, 1, 2, 3, 4, 5	6
	Z	6	$A_1A_3B_1B_2B_3A_2$	7, 0, 1, 2, 3, 4	5
g_8, gross, $P = 8$	X	7	$A_2A_1B_2B_3B_1A_3$	0, 1, 2, 3, 4, 5	6
	Z	0	$A_3B_1B_3B_2A_1A_2$	1, 2, 3, 4, 5, 6	7
g_9, gross, $P = 9$	X	8	$A_1B_1A_2A_3B_2B_3$	0, 1, 2, 3, 4, 6	7
	Z	7	$B_3B_2A_3A_2B_1A_1$	8, 1, 2, 3, 4, 5	6
u_{11}, gross, $P = 11$	X	0	$A_2A_1A_3B_3B_2B_1$	1, 2, 3, 4, 5, 9	10
	Z	10	$B_1B_2B_3A_3A_1A_2$	0, 4, 5, 6, 7, 8	9
IBM $[[90, 8, 10]]$, $P = 8$	X	7	$A_2B_2B_1B_3A_1A_3$	0, 1, 2, 3, 4, 5	6
	Z	6	$A_1A_3B_1B_2B_3A_2$	7, 0, 1, 2, 3, 4	5
b_9, $[[90, 8, 10]]$, $P = 9$	X	8	$A_1A_2B_2B_3A_3B_1$	0, 2, 3, 4, 5, 6	7
	Z	0	$B_1A_3B_2B_3A_2A_1$	1, 2, 3, 4, 5, 7	8
b_{10}, $[[90, 8, 10]]$, $P = 10$	X	9	$A_1A_2B_2A_3B_3B_1$	0, 1, 2, 3, 4, 7	8
	Z	8	$B_1B_3A_3B_2A_1A_2$	9, 2, 3, 4, 5, 6	7

TABLE S5. Attaining circuits: extended-code distance (Z/X basis), the lists used by Theorem 4 (per basis, "+f": with at most one readout flip), the direct search on the detector error model (no undetected logical of weight $< d_{\text{circ}}$ for the listed R, both bases), and the upper bound.

circuit	d_{ext}	lists, none realised in the bulk	direct search	upper bound	d_{circ}
g_8	9/9	w_9 432 (+f), w_{10} 17,136	$R \leq 3$	$R = 1$ witness, persistence	11
g_9	9/9	w_9 360 (+f), w_{10} 13,788	$R \leq 3$	$R = 1$ witness, persistence	11
g_{10}	10/10	w_{10} 3,204 (+f), w_{11} 121,104	$R \leq 3$	weight-12 data logical	12
u_{11}	10/10	w_{10} 144 (+f), w_{11} 7,344	$R \leq 2$	weight-12 data logical	12
IBM gross	9/8	Z: w_9 360; X: w_8 288 (+f), w_9 7,728	$R \leq 3$	$R = 1$ witness, persistence	10
IBM $[[90, 8, 10]]$	7/6	Z: w_7 315; X: w_6 270 (+f), w_7 4,320	$R \leq 4$	$R = 1$ witness, persistence	8
b_9	7/7	w_7 135 (+f), w_8 5,175	$R \leq 4$	$R = 1$ witness, persistence	9
b_{10}	7/7	w_7 225 (+f), w_8 5,535	$R \leq 4$	$R = 1$ witness, persistence	9
c_{11}	8/8	w_8 585 (+f), w_9 16,830	$R \leq 12$	weight-10 data logical	10

in both. As positive controls the same solvers find realisations of weight t (one more readout flip) for g_8, g_9, b_9, b_{10} and IBM's $[[90, 8, 10]]$ circuit.

Direct search. The search works on the detector error model of the simulator, not on the extended code: it branches on the mechanisms covering an odd detector, excludes earlier siblings, splits root cases by translation orbits (the observable action of each translation is solved and verified on all mechanisms) and closes the last levels with a meet-in-the-middle table. A minimum-weight logical has no proper undetected subset, so the absence of solutions up to weight W means $d_{\text{circ}} > W$ at that R. It also finds the $R = 1$ witnesses.

Upper bounds. For g_8, g_9, b_9, b_{10} and IBM's circuits the $R = 1$ witnesses persist (End Matter): the start offsets do not depend on R, and instance 1 of every reading check reads after the last layer of the one-round circuit. Replaying the witnesses of g_9, b_9, b_{10} and IBM's circuits in stim for R up to 30, at the start and shifted by two rounds, triggers no detector and flips the observable, while removing any one fault triggers a detector. For g_8, weight-11 undetected logicals found independently at R = 1–6, 8, 10, 12, 16, 20 and 30 were verified mechanism by mechanism in both bases.

S9. CERTIFICATES

VeriPB [24, 25] proofs are accepted only with the verdict "s VERIFIED UNSATISFIABLE"; DRAT proofs, produced by CaDiCaL [22], are checked with drat-trim [23] and their LRAT cores with the formally verified checker cake_lpr [26].

Lift instances are combined into batches by a selector disjunction, so that unsatisfiability of a batch implies that of each member.

Impossibility results. Five DRAT proofs (S7); regenerated formulas are byte-identical and re-verified.

g_{10}. Four VeriPB proofs (both sides, $d_{\text{ext}} \geq 10$ and completeness of the weight-10 and weight-11 lists); the opb inputs contain only parity, weight and blocking constraints, and translation symmetry enters only through `red` steps checked by VeriPB. 250 batches covering all $2 \times 124{,}308$ lift instances at $R^* = 13$, each refuted by CaDiCaL with a DRAT proof verified by drat-trim and an LRAT core verified by cake_lpr. The compression lemma (End Matter) rests on two computed premises: every fault spans at most one index in the round labelling of the memory circuit, and every bulk fault with indices in $[1, 12]$ exists in the circuit with 13 rounds with equal finite and bulk indices.

c_{11} *and* u_{11}. c_{11} shares its extended code with an earlier period-16 circuit (S13); $d_{\text{ext}} \geq 8$ and the completeness of the lists of 585 and 16,830 logicals are certified by VeriPB, and all lift instances for $R \leq 25$, both bases, by DRAT. For u_{11}, $d_{\text{ext}} \geq 10$ and the completeness of the lists of 144 and 7,344 logicals are certified by VeriPB 3, and all lift instances for $R \leq 31$ by DRAT. The largest proofs are generated deterministically and checked as streams; we record the SHA-256 hash of each so that it can be regenerated and compared.

The encodings were generated from the circuits by our code and cross-checked against independent implementations; the reductions (the staircase model, the criterion, the compression lemma, the gap argument and the persistence argument) are mathematical arguments whose premises were verified computationally. The other circuits of Table I are not certified; they rest on two independent solvers and the direct search.

IBM's depth-8 gross circuit in other fault models. For the CZ-compiled, SI1000-noise versions of IBM's gross-code circuit distributed with the Tesseract decoder [4] ($r = 12$, both memory bases), the circuit distance is 10: after projection onto the memory-basis detectors, which can only lower the distance, a VeriPB-checked certificate excludes undetectable logicals of weight ≤ 9, a weight-10 undetectable logical is verified on the unprojected circuit, and the projected detector error model coincides, up to a relabelling of checks, with our SD6 model of the same schedule at $R = 12$. In our re-implementation of the single-fault model of IBM's decoder (R noisy cycles followed by two noiseless ones) [1, 5], the circuit distance is 10 for the gross code and 8 for $[[108, 8, 10]]$ for every $R \geq 1$ in both bases; a written time-translation argument, whose premises we check by exact computation, reduces arbitrary R to the certified range $R \leq 11$ (resp. $R \leq 9$).

S10. LOGICAL ERROR RATES

Setup. All circuits are noised by the same SD6 implementation with $p_{\text{idle}} = \lambda p$; both bases are sampled separately and added. BP-OSD (min-sum, 10,000 iterations, OSD-CS order 7) and Relay-BP use identical parameters and exact priors.

Three tiers. (i) Exact, without statistical error: per round, IBM's gross circuit has 144 (Z) and 288 (X) undetected logicals of weight 10, IBM's $[[90, 8, 10]]$ circuit 315 and 540 of weight 8, c_{11} 24,507 of weight 10 in each basis. These counts are structural information only: the undetected terms they give lie many orders of magnitude below the measured rates. (ii) Measured, by direct sampling or by stratified sampling at fixed fault number k requiring at most monotonicity of the failure probability f_k in k: at $\lambda = 1$, $p = 2 \times 10^{-3}$, BP-OSD, g_{10} and u_{11} have per-round rates 1.45×10^{-4} $[0.81, 2.40]\times 10^{-4}$ and 3.39×10^{-4} $[2.34, 4.76]\times 10^{-4}$, about 5–10 times IBM's; the ratio to IBM stays above about 1.5 down to 1.5×10^{-3} for every smooth form of f_k we tried; c_{11} is worse than IBM's $[[90, 8, 10]]$ circuit at all $p \geq 10^{-3}$ (monotonicity bound). At $\lambda = 0$, c_{11} over IBM is 0.51 [0.31, 0.83] at 1.2×10^{-3} (direct counts, 32 failures in 3.8×10^6 and 33 in 2×10^6 shots) and below 1 at every sampled p in 2–5 $\times 10^{-3}$ with both decoders (0.52 [0.44, 0.62] at 3×10^{-3}, pooled from two independent pipelines); at $\lambda = 0.1$ the two are statistically indistinguishable, and at $\lambda = 1$ c_{11} is worse by 6–9 (BP-OSD) and 9–17 (Relay-BP) at 1.5–5 $\times 10^{-3}$. (iii) Extrapolated: everything below about 10^{-3}. A power law for f_k and a threshold form $f_k = 1 - (1-\theta)^{\binom{k}{t}}$ fit equally well but place the crossover with IBM between about 10^{-3} and below 10^{-4}, or exclude it, depending on the number t of faults at which BP-OSD starts to fail, which the data do not resolve.

Decoder. Of 24 sampled failures (c_{11} and IBM's $[[90, 8, 10]]$ circuit, $\lambda = 1$, $k = 16$), BP-OSD returned a correction less likely than the actual fault set in 23; a higher OSD order changed none of them.

S11. LOGICAL ERROR RATES IN A NEUTRAL-ATOM NOISE MODEL

Architecture and timing. Data atoms stay in the entangling zone; the X- and Z-ancilla arrays are held in movable tweezers and translated rigidly, and each translation is followed by a global Rydberg CZ pulse and local single-qubit

gates (CX = H·CZ·H on the target). One move serves one pair (check type, polynomial term). Both codes have three terms in each of A and B, so every translation-invariant schedule needs exactly 12 moves per round; this holds for the period-11 circuits u_{11} and c_{11} studied here and for IBM's depth-8 circuits, which also have two mid-circuit readout layers per round. In the main model the two ancilla arrays move one after the other, as in qSIEVE [28], and each move takes a fixed time. Equal move counts do not imply equal durations: with a move time growing as the square root of the distance, the summed path is 6% longer for u_{11} and 21% longer for c_{11} (layout dependent); at current parameters this changes the fault mass by less than 2%.

Noise. Before each layer every atom receives a Z error with probability $(1-e^{-\tau/T_2})/2$, with τ the move time plus the readout wait; during a mid-circuit readout the unmeasured atoms receive the scattering error of Table S6; preparation, gates, measurement and the two ancilla transfers per round carry the errors of Table S6. The set "cur×0.5" halves every error probability of "cur"; it is a uniform scaling, not a literature value.

TABLE S6. Parameters of the neutral-atom model (error probabilities unless stated). "cur": current hardware; "proj": projected.

quantity	cur	proj	sources and remarks
CZ	5.0×10^{-3}	3.0×10^{-3}	cur: 99.5% [29], [30, 31]; proj: 99.72%, 99.73% with loss excluded [32, 33], and 99.71% [34] (leakage-corrected, from a different architecture)
single-qubit gate	9×10^{-4}	3×10^{-4}	cur: local gates, 99.912% [30], 99.902% [33]; proj: global Raman estimate 99.97% [29], applied here to local gates
preparation	6.8×10^{-3}	4×10^{-3}	99.32% [30]; 99.6% [29]
measurement	5.8×10^{-3}	2×10^{-3}	0.46% flips plus half of 0.24% loss [31]; imaging ≈99.8% [30]
transfer	1×10^{-3}	1×10^{-3}	$\lesssim$0.1% per transfer [30]
T_2	1 s	10 s	cur [30, 31]; proj is an assumption, as in [15, 28]
time per move	200 μs	200 μs	[30]
readout wait	1 ms	0	cur [30]; proj: pipelined with the next moves [15], assumes a second ancilla set
readout scattering	6.8×10^{-4}	3.4×10^{-4}	0.034% per 500 μs image [30], applied for 1 ms

Sampling. The $[[90, 8, 10]]$ circuits are sampled for $R = 10$, both bases separately, with BP-OSD and Relay-BP. Shot numbers were fixed in advance for each block; no block was extended after its failures were seen. An independent implementation of the model gives identical detector error models and reproduces the ratios.

TABLE S7. Per-round logical error rate ratio, c_{11} over IBM's depth-8 $[[90, 8, 10]]$ circuit, with 95% intervals; failures summed over both bases (c_{11} / IBM) and shots per basis. Main model (sequential moves, no error on unpaired atoms).

set	decoder	ratio	failures	shots
cur	BP-OSD	0.82 [0.69, 0.97]	232 / 275	640
cur	Relay-BP	0.72 [0.55, 0.95]	87 / 119	530
cur×0.5	BP-OSD	0.54 [0.42, 0.69]	93 / 173	21000
cur×0.5	Relay-BP	0.40 [0.28, 0.56]	44 / 111	48000
proj	BP-OSD	0.57 [0.46, 0.71]	125 / 219	25000
proj	Relay-BP	0.48 [0.35, 0.66]	53 / 111	41000

Sensitivity. With simultaneous moves the advantage is not significant at current parameters (0.89 [0.76, 1.04]) but persists at projected gate errors (0.61 [0.49, 0.76]); with the measured round timing of Ref. [31] it is 0.92 [0.80, 1.06]. If in addition to simultaneous moves unpaired atoms suffer half the CZ error, the ordering reverses (2.41 [2.09, 2.77] at current parameters, BP-OSD), because c_{11} spreads its CNOTs over more pulses and leaves more atoms unpaired (630 vs. 90 unpaired atom–pulse exposures per round). Experiments report no error on atoms parked in a separate storage zone [30, 31]; we know of no measurement for unpaired atoms inside the entangling zone. Atom loss is folded into Pauli errors and the decoders receive no erasure information; T_1 decay and heating are not modelled. The gross circuits are not reported in this model.

S12. A TRANSFER EXAMPLE

The timing template of u_{11} (Table S1) is legal and deterministic for every BB code with three terms in A and B. Applied to $[[108, 8, 10]]$ it gives a period-11 circuit with $d_{\text{ext}} = 8$ and lists of 54 and 2,268 extended logicals of weights 8 and 9 per basis, none realised in the bulk; the window conditions hold, so $d_{\text{circ}} = 10$ for every R in both bases (a direct search finds no undetected logical of weight ≤ 9 at $R = 1$, 2 and one of weight 10 at $R = 1$). A depth-8 circuit with $d_{\text{circ}} = 10$ is already known for this code, so the example only shows how the criterion transfers. On $[[90, 8, 10]]$ the same template, with the same term labels, realises 45 weight-9 logicals in the bulk; c_{11} is the template after exchanging the two blocks and relabelling the terms.

S13. EARLIER CONSTRUCTIONS

Non-interleaved bound for $[[90, 8, 10]]$. Up to swaps of positions 1,2 and 5,6 and reversal, a check of this code has 90 order types; for 36 of them the extended code of that check alone has a logical of weight ≤ 8. For the remaining 54, any realised residual columns inside one of the 558 weight-10 X-type logicals whose total size reduction is at least 2 give an extended logical of weight ≤ 8. A SAT encoding of the choice of one type per check with one clause per such combination (17,910 combinations, each verified) is unsatisfiable, with a DRAT proof verified by drat-trim. Hence every assignment of orders has $d_{\text{ext}} \leq 8$ on the X-check side and every non-interleaved circuit has $d_{\text{circ}} \leq 8$ by SBB's Cor. 1.

Longer periods. Before the class $\mathcal{T}$ was analysed, full distance was reached at longer periods: for $[[90, 8, 10]]$ by a translation-invariant period-16 circuit with three CNOT-free steps (orders $A_2A_1A_3B_2B_3B_1$ for X checks and $B_1B_3B_2A_3A_1A_2$ for Z checks, with the extended code of c_{11}) and a non-wrapping period-12 circuit; for the gross code by colour-symmetric circuits of periods 18 and 21 whose orders lie in the class of SBB's Fig. 12, and therefore share its extended code ($d_{\text{ext}} = 11$, 4,224 weight-11 logicals per basis). All have $d_{\text{circ}} = d$ for every R; they are superseded by c_{11} and g_{10}.

SBB's Fig. 12. It is non-interleaved, so SBB's Prop. 3 and Cor. 1 give $d_{\text{circ}} = d_{\text{ext}}$ for every R, and two independent exhaustive searches give $d_{\text{ext}} = 11$; hence $d_{\text{circ}} = 11$, as SBB conjectured. It is colour-tiled rather than translation invariant and therefore not in $\mathcal{T}(8)$. Its orders cannot reach 12 by retiming within several symmetric families (bulk statements, each certified by DRAT); periodic schedules of period 8 keeping its orders are, up to translation, four non-interleaved circuits, each with weight-11 undetected errors for $R = 1, \ldots, 6$.

S14. VERIFICATION AND DATA

All circuits were built and simulated with Stim [21]; feasibility questions were decided with CP-SAT, CDCL SAT solvers and a Bellman–Ford solver, and extended-code searches use dedicated depth-first and meet-in-the-middle searches. Every result was reproduced by a separately written implementation: the class enumeration, the bulk model and its agreement with the detector error model, the pattern coverage, the lists, the bulk lifts and the direct searches. The data set contains the circuit specifications, the lists of extended logicals, the impossibility patterns, the upper-bound witnesses, the certificates (formulas, proofs up to 100 MB, and SHA-256 hashes with regeneration scripts for larger ones) and the reproduction scripts; it is available at [Zenodo DOI].